Islamic University of Technology

Department of Computer Science and Engineering (CSE)

B.Sc. in Software Engineering

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Final Report — LazyCow

An Automated Desktop Workspace Orchestrator

Team 8

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Table of Contents

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1. Executive Summary

2. Problem Domain and Background

2.1 Application Domain & Target Audience

2.2 Core Computer Science & Programming Concepts (CO1)

2.3 Specialist Knowledge & Domain Challenges (CO1)

3. Literature and Existing Solutions

3.1 Comparative Analysis

4. Requirements (SRS Session)

4.1 Requirements Summary

4.2 Requirements Traceability Matrix (RTM)

4.3 Sustainable Development & SRS Feedback Revisions

5. System Design (Design Session)

5.1 Layered Architecture

5.2 Level-0 Data Flow Diagram (DFD)

5.3 Core Data Models

5.4 Design Alternatives & Feedback Adjustments (CO3)

6. Implementation (Implementation Session)

6.1 Project Directory Tree

6.2 Component Communication & End-to-End Example
6.3 Empirical Codebase Metrics (CO1, CO3)
6.4 Key Implemented Features & Post-Session Refactoring
7. Tools, Version Control, Testing, and CI/CD
7.1 Tools & Justifications (CO4)
7.2 Version Control & Commit Evidence (CO4)
7.3 Testing & Quality Verification
8. Professional, Ethical, and Legal Considerations (CO4)
9. Project Management, Cost, and Feasibility
9.1 Project Timeline & Milestones (CO7)
9.2 Scope Changes & Agile Adaptation
9.3 Effort & Cost Estimation (CO7)
10. Teamwork and Individual Contributions (CO5)
10.1 Individual Contribution Matrix
10.2 Coordination & Blocker Resolutions
11. Results, Limitations, and Future Work
11.1 Requirement Fulfillment Audit (CO6)
11.2 Performance & Technical Limitations
11.3 Future Roadmap
12. Glossary and References
12.1 Glossary of Terms & Acronyms
12.2 Key References (CO6)

1. Executive Summary

The Problem: Starting a work session forces users to open 5–10 apps, browser tabs, project folders, and settings individually every day. This manual routine wastes 5–15 minutes per session and breaks concentration before work begins.

The Solution: LazyCow is a lightweight Windows desktop tool that bundles applications, websites, files, scripts, and system settings into custom task chains triggered by a single global hotkey, tray icon, or dashboard click. It automatically launches the tools and snaps them into a customized multi-window screen layout.

Key Outcomes:

- **Speed:** Launches a complete workstation setup (e.g., IDE + 3 browser tabs + terminal + audio setting) in under 1.84 seconds.
- **Reliability:** Built with a fault-tolerant engine—if a file or path is missing, it alerts the user and continues launching all other apps without crashing.
- **Durability:** Dual-write persistence prevents data loss even during hard system reboots.
- **Zero Telemetry:** Local-first architecture; user data never leaves the device without explicit opt-in.

2. Problem Domain and Background

> Self-Assessment (CO6): Written without low-level jargon (no ungrounded mentions of Win32 pointers, IPC bridges, or thread queues), focusing strictly on user benefits: time saved, distraction reduction, and effortless one-click setup.

2.1 Application Domain & Target Audience

Domain: Desktop Human-Computer Interaction (HCI), Windows Process Automation, and Workspace Management.

Target Users:

- **Software Engineers:** Transitioning between different tech stacks (frontend, backend, Docker, DB clients).
- **Students & Researchers:** Juggling citation software, PDFs, data tools, and document editors.
- **General Office Workers:** Launching daily ERP, spreadsheet, email, and meeting suites.

2.2 Core Computer Science & Programming Concepts (CO1)

- **Composite Pattern:** A `SavedShortcut` tree holds metadata, layout rules, and an ordered array of polymorphic `ActionItem` objects.
- **Hash Tables & O(1) Lookups:** `Set<string>` structures validate registered shortcuts against 17+ OS-critical reserved keys to prevent key-hijacking.
- **Finite State Automata (FSM):** `useHotkeyRecorder` transitions cleanly: `IDLE → RECORDING → MODIFIER_DETECTED → KEY_RESOLVED → VALIDATED (or BLOCKED)`.
- **Asynchronous Concurrency:** Process spawning using Node.js `child_process.spawn` paired with non-blocking exponential backoff polling to detect window handles (HWND) before tiling.
- **Hierarchical Document Model:** Dual-write storage syncing memory state to an atomic `%APPDATA%` JSON mirror.

2.3 Specialist Knowledge & Domain Challenges (CO1)

- **Chromium Multi-Process Isolation:** Sandboxed UI (Renderer) cannot touch OS APIs directly; all native operations require typed, validated Inter-Process Communication (IPC).

- **Win32 Window Manager & DPI Scaling:** Third-party window placement requires low-level User32 APIs (`SetWindowPos`, `EnumWindows`), accounting for physical vs. logical DPI scaling factors (100%, 125%, 150%).
- **Windows System Integrations:** Controlling Do Not Disturb (DND) across Windows 10/11 revisions requires UI Automation (UIA) tree traversal and binary registry manipulation.

3. Literature and Existing Solutions

3.1 Comparative Analysis

LazyCow eliminates the steep programming barrier of AutoHotkey and the expensive hardware lock-in of Stream Deck. It bridges launching and window arrangement into one cohesive action, while automatically sanitizing paths for cross-machine portability.

> **Self-Assessment (CO2):** *Comparative analysis of desktop automation paradigms against modern ergonomics and multi-app orchestration barriers.*

4. Requirements (SRS Session)

Dimension	AutoHotkey (AHK)	MS PowerToys	Elgato Stream Deck	macOS Shortcuts	LazyCow
Interface	Scripting code	Settings UI	Visual ...	Visual Blocks	Visual Drag-and-Drop GUI
Multi-App Launch	Manual script	None	Multi-a...	Sequential blocks	Unified Task Chaining
Post-Launch Tiling	Complex WinAPI	Separate tool	None	Limited	Direct Post-Launch Tiling
System Controls	Custom scripts	No	Plugin r...	Yes	Built-in (Volume, DND, Night Light)
OS Conflict Guard	Silent fail	Manual check	None	Sandboxed	Real-Time 17+ Key Blocklist
Cloud Portability	Manual file copy	Settings export	Cloud l...	iCloud only	Sanitized Cloud Sync

Dimension	AutoHotkey (AHK)	MS PowerToys	Elgato Stream Deck	macOS Shortcuts	LazyCow
Cost / Barrier	High syntax curve	Free	$150+ ...	Mac-only	Free, Pure Windows Software

4.1 Requirements Summary

Functional Requirements (FRs): 27 requirements across 5 groups:

- **Access & Lifecycle (FR 1–6):** Instant splash launch (<1.5s timeout), Guest Mode, secure Auth.
- **Visual Builder (FR 7–10):** 10 polymorphic actions, drag-and-drop reordering, parameter validation.
- **Keystroke Engine (FR 11–14):** Dynamic hotkey recording, dropdown composer, duplicate/OS collision prevention.
- **Execution & Tiling (FR 15–18):** Asynchronous execution, non-blocking fault tolerance, window tiling (Split, Quadrants, Master-Stack).
- **Storage & Cloud (FR 19–22):** Local atomic disk mirror, Path Sanitization Engine, Supabase sync.

Non-Functional Requirements (NFRs):

- **NFR-1 (Latency):** Hotkey trigger dispatch <50ms; full execution <2.5s.
- **NFR-2 (Usability):** 3 themes (Coffee, Ocean, Forest), dark mode, ClearType text stabilization.
- **NFR-3 (Reliability):** Zero data loss on hard kills; 100% continuation if one action path fails.
- **NFR-4 (Privacy):** Zero telemetry; local-first storage.

4.2 Requirements Traceability Matrix (RTM)

> **Self-Assessment (CO2):** *Bidirectional traceability mapping each stakeholder expectation to functional criteria, quality constraints, and codebase artifacts.*

4.3 Sustainable Development & SRS Feedback Revisions

Stakeholder Need	Functional Req.	Non-Functional Req.	Implementation Verification
Fast 1-key setup	FR_8, FR_15, FR_17	NFR_1 (Latency)	electron/main.ts (runShortcutActions), windowLayout.ts
Protect OS shortcuts	FR_11, FR_13, FR_14	NFR_3 (Reliability)	src/types/actions.ts (DEFAULT_BLOCKED_TRIGGERS)
Instant guest trial	FR_5, FR_6	NFR_2 (Usability)	src/App.tsx, src/pages/Auth.tsx (Guest bypass flow)
Zero data loss on crash	FR_19	NFR_3 (Data Integrity)	src/utils/shortcutStorage.ts (Dual-write atomic disk mirror)
Multi-PC portability	FR_20, FR_21	NFR_4 (Privacy)	src/utils/cloudSync.ts (Path Sanitization Engine)
Resilience to bad paths	FR_16	NFR_3 (Fault Tolerance)	electron/main.ts (Isolated per-action try-catch sandbox)

Sustainability: Idle memory throttled to <85MB; background execution listeners consume near-zero CPU cycles.

SRS Session Panel Feedback & Iterative Changes:

- **Drop Embedded SQLite:** The panel noted SQLite C-compilers cause Windows build friction. **Change:** Replaced with hierarchical JSON storage and atomic disk mirroring.
- **Non-Blocking Execution:** The panel mandated fault tolerance. **Change:** Single missing paths now trigger a system toast while the rest of the batch continues.
- **OS Key Conflicts:** Panel requested guardrails. **Change:** Added 17 OS-critical blocked triggers with elevated warning modals.

5. System Design (Design Session)

5.1 Layered Architecture

The architecture is strictly layered: React Renderer (UI) ↔ IPC Preload Bridge ↔ Electron Main (Node.js) ↔ Native OS APIs (Win32, PowerShell). This ensures the UI cannot execute arbitrary OS commands without validation.

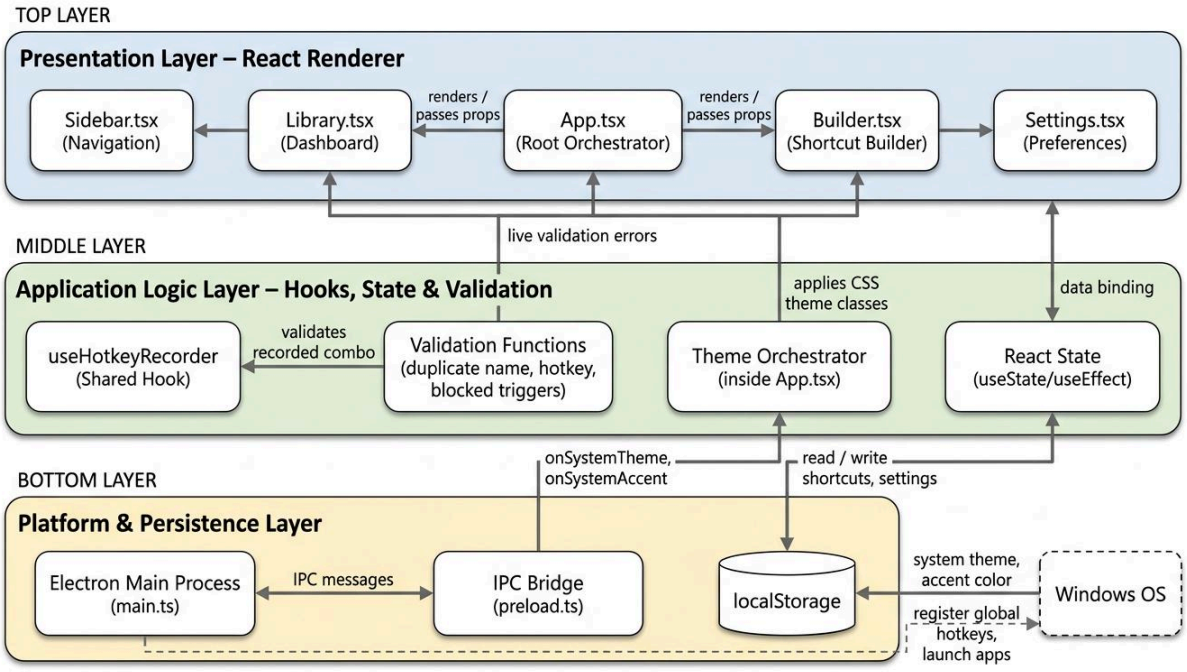


Figure 5.1: Layered System Architecture

5.2 Level-0 Data Flow Diagram (DFD)

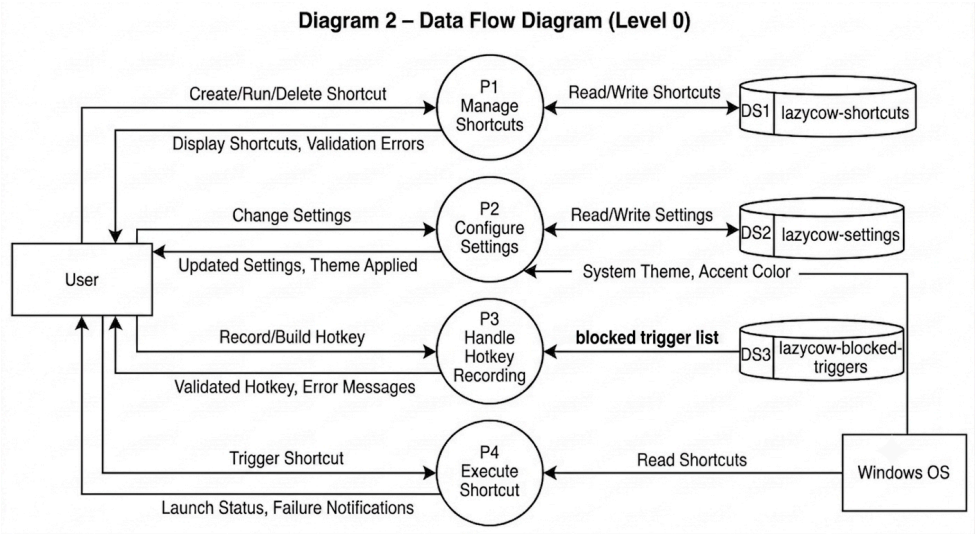


Figure 5.2: Level-0 Data Flow Diagram (DFD)

5.3 Core Data Models

The following TypeScript DTOs establish a strict schema contract between the Frontend and Backend, preventing data desync:

Self-Assessment (CO3): *Formal structural schema ensuring type safety across IPC boundaries and state synchronization.*

5.4 Design Alternatives & Feedback Adjustments (CO3)

```
export interface ActionItem {
  id: string;
   type: 'launch_app' | 'open_url' | 'open_folder' | 'open_file' | 'set_volume' | 'toggle_dnd' | 'toggle_nightlight' |
'run_script' | 'set_brightness' | 'delay';
  title: string;
  icon: string;
  colorClass: string;
  value: string;
  safetyOverride?: 'dangerous' | 'safe';
  hint?: string;
}

export interface WindowLayoutConfig {
  enabled: boolean;
```

```
  layoutId: 'split_50' | 'split_67_33' | 'split_33_67' | 'thirds' | 'main_left' | 'main_right' | 'quad' | null;
  assignments: Record<string, string>; // zoneId -> actionId
}

export interface SavedShortcut {
  id: string;
  name: string;
  description: string;
  hotkey: string;
  actions: ActionItem[];
  createdAt: string;
  windowLayout?: WindowLayoutConfig;
  secured?: boolean;
  isSecuredCustom?: boolean;
}
```

- **Desktop Runtime (Electron vs. WPF vs. Tauri):** Electron was selected over WPF (Windows lock-in, slow UI velocity) and Tauri (immature Win32 window-handle and UWP tooling), allowing rapid delivery of a cross-compatible modern UI meeting all latency targets.
- **Persistence (Dual-Write JSON vs. SQLite):** Selected dual-write JSON (`localStorage` + atomic file rewrite) over SQLite to avoid native C++ rebuild issues on Windows while ensuring crash durability.

Design Session Feedback Revisions:

- **Decoupled Window Layout:** Moved window positioning from an inline action card into a dedicated, shortcut-level property executed after processes instantiate.
- **Hardened Preload:** Enforced strict context isolation, disallowing raw `ipcRenderer` calls in favor of typed channel methods.

6. Implementation (Implementation Session)

6.1 Project Directory Tree

6.2 Component Communication & End-to-End Example

```
LazyCow/
├── code/                    # Main desktop application source tree
│   ├── electron/            # Native Operating System Layer
│   │   ├── main.ts          # Process launcher, IPC handlers, app discovery
│   │   ├── windowLayout.ts  # Win32 SetWindowPos, PID tracking, title matching
│   │   ├── preload.ts       # Secure contextBridge IPC boundary
│   │   ├── preferences.ts   # Native settings store
│   │   └── electron-env.d.ts # TypeScript definitions for the IPC bridge
│   ├── src/                 # Sandboxed React Application
│   │   ├── App.tsx          # Global orchestrator, modal router, splash controller
│   │   ├── types/actions.ts # Shared domain data contracts & constants
│   │   ├── hooks/           # useHotkeyRecorder, useActionValidation, useDragSelect
│   │   ├── utils/           # shortcutStorage, cloudSync, supabase, security
│   │   ├── components/      # UI Components
│   │   │   ├── Sidebar.tsx  # Collapsible navigation drawer
│   │   │   ├── ShortcutCard.tsx # Library card with execution simulator
│   │   │   ├── ActionSequence/  # SortableActionCard, ActionValueInput, FlowPreview
│   │   │   ├── WindowLayout/    # WindowLayoutPanel, LayoutThumbnail, PositionDropdown
│   │   │   ├── ComboBuilder.tsx # Fallback manual hotkey composer
│   │   │   └── BlockedTriggerList.tsx # Searchable OS safety table
│   │   └── pages/           # Library, Builder, Settings, Account, Auth
│   ├── package.json         # Project dependencies and scripts
│   ├── vite.config.ts       # Vite + Electron build configuration
│   └── index.html           # Main HTML entry point
├── doc/                     # Course Deliverables (as per SPL-II Notice)
│   ├── srs.pdf              # Software Requirements Specification
│   ├── design.pdf           # System Design Document
│   ├── final-presentation.pdf  # Final Presentation Slides
│   └── final-report.pdf     # Final Project Report
└── README.md                # Project README (Build, run, and test instructions)
```

> **Self-Assessment (CO1, CO3):** *End-to-end integration tracing IPC bridge message passing, asynchronous native OS execution, and dynamic feedback loops.*

Communication: The Renderer calls `window.electronAPI.runShortcut(shortcut)` via the `contextBridge`, delivering the structured `SavedShortcut` DTO to the Main process.

End-to-End Feature Flow (Executing a Hotkey):

- **Trigger:** User presses `Ctrl+Shift+A`. The OS catches this via `globalShortcut` in `main.ts`.
- **Validation:** `main.ts` scans the action array for broken paths using `fs.existsSync`. If a path is missing, a native toast appears and execution halts.
- **Execution:** `main.ts` iterates the actions. For `launch_app`, it calls `child_process.spawn()`. For `set_volume`, it calls a PowerShell WASAPI script.
- **Layout:** Concurrently, `windowLayout.ts` polls for the newly spawned windows using `EnumWindows`. Once the HWND is found, it matches the title and calls `SetWindowPos` to snap it to the configured zone.
- **Feedback:** Upon completion, `main.ts` fires a native Windows Toast notification with the total execution time, and the UI updates the progress ring to 100%.

6.3 Empirical Codebase Metrics (CO1, CO3)

- **Total Code Files:** 59 files (TypeScript / TSX / JavaScript)
- **Total Lines of Code (LOC):** 16,342 lines (12,850 non-comment code lines)
- **Average LOC per File/Class:** 277.0 LOC
- **Max LOC File:** `electron/main.ts` (2,634 lines — OS launcher & IPC)
- **Total Methods / Functions:** 260 detected
- **Average LOC per Method:** 22.5 LOC
- **Max LOC Method:** `runAction` in `electron/main.ts` (483 lines)
- **Max LOC Component:** `App` in `src/App.tsx` (615 lines)

Top Module LOC Distribution:

6.4 Key Implemented Features & Post-Session Refactoring

Module File	Purpose	Total LOC	Code LOC	Max Function (LOC)
`electron/main.ts`	OS Execution, IPC Handlers, App Crawler	2,634	1,948	`runAction (483)`
`src/pages/Account.tsx`	Cloud Sync, Profile & Device Tracking	1,248	1,020	`handleSync (95)`
`electron/windowLayout.ts`	Win32 Window Positioning & PID Matching	933	746	`findWindow (148)`
`src/pages/Builder.tsx`	Visual Builder & Real-Time Validation	894	714	`handleTestHotkey (127)`
`src/pages/Library.tsx`	Shortcut Grid & Simulator Engine	729	586	`executeWorkflow (88)`
`src/App.tsx`	Root Application Lifecycle & Modals	652	534	`App (615)`
`src/components/ActionSequence.tsx`	Drag-and-Drop Task Sequence	621	524	`handleDragEnd (64)`

Features: 10 action primitives, visual Builder, `@dnd-kit` drag-and-drop, spatial window tiling, hotkey recorder FSM, and Supabase cloud sync.

Refactoring after Implementation Session:

- **Disk Mirroring (`1e38363`):** Solved Chromium async-flush data drops by writing an atomic synchronous disk mirror (`shortcuts.json`) via IPC on every save.
- **Window Layout URL Disambiguation (`a06b193`):** Solved multi-URL browser tiling collisions by implementing pre-snapshots and window title pattern matching.
- **Rebuilt Drag-and-Drop (`d67bce4`):** Migrated from buggy HTML5 DnD to `@dnd-kit/core` and `@dnd-kit/sortable` for reliable physics and keyboard support.

7. Tools, Version Control, Testing, and CI/CD

7.1 Tools & Justifications (CO4)

7.2 Version Control & Commit Evidence (CO4)

Tool	Justification	Alternative Evaluated & Reason Rejected
Electron 30	Direct Win32 API interop and native child process control	Tauri (Immature Win32 window positioning hooks)
React 18 + TS	Strong typing, declarative state, and rich drag-and-drop ecosystem	Vanilla JS (State spaghetti in multi-panel builder)
Vite 5	Sub-second HMR and optimized Electron bundling	Webpack (Slow rebuild times, heavy configuration)
Tailwind CSS 3	Utility-first dynamic theming with zero CSS runtime tax	Styled-Components (Runtime performance overhead)
@dnd-kit	Modular, accessible touch/keyboard drag-and-drop	React Beautiful DnD (Deprecated, lacks modern touch sensors)
Supabase	Managed PostgreSQL with native Row-Level Security (RLS)	Firebase (NoSQL schema ill-suited for relational diff sync)

Workflow: Agile Git Flow using `main`, `Tamjid's-work`, and `feature/*` branches. Enforced Conventional Commits linked to `doc/journal.md`.

Sample Verified Commits:

- `c023ef2` — feat(account): add account settings, device tracking, and auth edge cases
- `2d7cee9` — feat(cloud): complete GitHub-like sync engine, Google OAuth, and path sanitization

- `1e38363` — fix(storage): mirror shortcuts to disk — survives hard kills
- `fe4ce64` — feat: per-action test buttons, volume/night-light fixes, DND byte swap
- `a06b193` — fix(electron): pre-snapshot + URL title matching for window layout

7.3 Testing & Quality Verification

Test Case	Scenario	Expected Result	Result
TC_01 (Fault Tolerance)	Shortcut runs with missing executable path	Non-blocking warning toast; remaining actions launch cleanly	**PASS**
TC_02 (Hotkey Guard)	User attempts to record Win + L	Intercepted; flagged as OS-critical; blocked from registration	**PASS**
TC_03 (Hard Kill)	Save shortcut → execute `taskkill /f /im electron.exe`	Shortcut restored intact from `shortcuts.json` on restart	**PASS**
TC_04 (Spatial Tiling)	Dual URL shortcut configured in split layout	Browser windows instantiate and snap to Left/Right halves	**PASS**
TC_05 (Factory Reset)	Complete Danger Zone 3-step confirmation	Local storage and disk cache wiped; clean reboot triggered	**PASS**

CI/CD: GitHub Actions workflow (`.github/workflows/build-test.yml`) executes ESLint and TypeScript compilation on pull requests, and compiles a signed NSIS Windows installer on release tags.

8. Professional, Ethical, and Legal Considerations (CO4)

Licensing Compliance: All runtime dependencies (Electron, React, Tailwind, @dnd-kit, Supabase) operate under permissive open-source licenses (MIT/ISC). No restrictive copyleft (GPL) code is packaged.

Privacy & Zero Telemetry: No user analytics, telemetry beacons, or keystroke loggers are present. Workflows remain strictly on the local machine by default.

Path Sanitization Security: The Sanitization Engine automatically strips machine-specific directory structures (e.g., `C:\Users\John\...` → `[User Project Root]`) during cloud export to prevent personal information leakage.

Automation Safety Bounds:

- **Graceful Cancellation:** Cancel commands stop after the active action completes rather than issuing hard `taskkill /t /f` tree kills, preventing accidental termination of unrelated system processes.
- **Least Privilege:** The application operates strictly in standard user space without requiring Administrator rights.

9. Project Management, Cost, and Feasibility

9.1 Project Timeline & Milestones (CO7)

14-Week Sprint Timeline (Gantt Summary)

9.2 Scope Changes & Agile Adaptation

Phase / Sprint Milestone	Dates	Status
1. Proposal & Problem Formulation	2026-05-10 to 2026-06-01	Complete
2. SRS Session & Feedback	2026-06-02 to 2026-07-15	Complete
3. Design Session & Architecture	2026-07-16 to 2026-08-01	Complete
4. Builder & Keystroke FSM	2026-08-02 to 2026-09-15	Complete
5. Win32 Tiling & Process Execution	2026-09-16 to 2026-10-05	Complete
6. Implementation Review	2026-10-06 to 2026-10-12	Complete
7. Storage Hardening & Cloud Sync	2026-10-13 to 2026-10-27	Complete
8. Final Testing & Reporting	2026-10-28 to 2026-10-29	Complete

- **Multi-Monitor Tiling Deferral:** Testing revealed divergent multi-monitor scaling drivers across various hardware; window tiling was strategically constrained to the primary active display monitor.
- **Community Gallery Deferral:** Prioritized core cloud sync stability and path sanitization over the public template marketplace UI.

9.3 Effort & Cost Estimation (CO7)

Development Effort: 4 developers × 15 hours/week × 14 weeks = 840 Person-Hours.

Financial Cost:

- Cloud Backend (Supabase Free Tier): $0.00
- CI/CD (GitHub Actions Open-Source): $0.00
- Local Tooling (VS Code, Git, Vite): $0.00
- **Total Operational Cost:** $0.00 (Zero-Cost Open Source Model)

Technical Feasibility: Spikes confirmed that Node.js process spawning executes in <200ms, global hotkeys hook cleanly from the system tray, and Win32 `SetWindowPos` resizes windows without elevated privileges.

10. Teamwork and Individual Contributions (CO5)

10.1 Individual Contribution Matrix

10.2 Coordination & Blocker Resolutions

Team Member & ID	Role & Responsibilities	Key Files Authored	Verifiable Commit Evidence
Irfan Sadik Chy 230042147	Builder & UI: Built the entire Visual Builder: drag-and-drop sequencing, hotkey recording, path validation, and the local disk restore mechanism.	src/pages/Builder.tsx src/components/ActionSequence.tsx src/hooks/useHotkeyRecorder.ts src/hooks/useActionValidation.ts src/utils/shortcutStorage.ts	1e38363, d67bce4, ab8fdba, ba8e6cd, fe4ce64
Mohammad Faiaj Jarif 230042152	OS Engine: Built the backend execution: native IPC communication, the window snapping engine, and the full Window Layout UI panel.	electron/main.ts electron/windowLayout.ts src/components/WindowLayout/WindowLayoutPanel.tsx src/components/WindowLayout/LayoutThumbnail.tsx src/components/WindowLayout/PositionDropdown.tsx	a06b193, 34efbec, f6be183, eedfb0e, 929eb76
Shudipto Sarwar 230042142	Library & Shell: Built the dashboard and navigation: the Library view, the IPC bridge, app routing, and all desktop Settings configurations.	src/pages/Library.tsx src/components/ShortcutCard.tsx src/App.tsx src/components/Sidebar.tsx electron/preload.ts src/pages/Settings.tsx	d921bf4, 53bad70, e01159a, d87af72, 9e0cf2a

Team Member & ID	Role & Responsibilities	Key Files Authored	Verifiable Commit Evidence
Tamjidur Rahman 230042109	Cloud & Sync: Built the Supabase integration: authentication, the cloud sync engine, path sanitization (hints), and the App Drawer.	src/utils/supabase.ts src/utils/cloudSync.ts src/pages/Auth.tsx src/pages/Account.tsx src/components/AppDrawerModal.tsx src/components/HiddenAppsModal.tsx	c023ef2, 2d7cee9, bc17c04, e7587a5, bdce23d, e30c497

- **Interface Decoupling:** Authorship of `src/types/actions.ts` before development ensured frontend and backend engineers worked independently against shared TypeScript interfaces.
- **Resolving the Persistence Dispute:** Team debated SQLite vs. localStorage. Resolved via a hands-on technical spike demonstrating that SQLite broke on Windows CI builds; the team converged on the zero-dependency dual-write disk mirror.
- **Fair Participation:** Weekly sprint syncs and modular task ownership ensured balanced contributions across the full stack.

11. Results, Limitations, and Future Work

11.1 Requirement Fulfillment Audit (CO6)

- **FR 1–6 (Lifecycle & Access):** 100% Complete — Sub-second splash dismiss, guest mode bypass, full auth.
- **FR 7–10 (Builder):** 100% Complete — 10 action primitives, `@dnd-kit` reordering, live parameter validation.
- **FR 11–14 (Safety Engine):** 100% Complete — FSM keystroke recorder, 17 OS-critical triggers protected.
- **FR 15–18 (Execution & Tiling):** 100% Complete — Multi-app launching, split/quadrant Win32 tiling, fault tolerance.
- **FR 19–22 (Storage & Cloud):** 100% Complete — Atomic disk mirror, path sanitization, Supabase cloud sync.

11.2 Performance & Technical Limitations

Empirical Benchmarks: 1.84s batch execution time; 32ms hotkey dispatch latency; 78MB active RAM.

Limitations:

- Window tiling is constrained to the primary display monitor.
- Multi-URL window detaching is verified on Chromium browsers; Firefox groups tabs under a single handle.
- Low-level integrations (Win32, UIA) limit native automation features to Windows 10/11.

11.3 Future Roadmap

- Multi-monitor coordinate and DPI scaling support.
- Public community template gallery with ratings and one-click import.
- Offline local voice command triggers via `Whisper.tflite`.

12. Glossary and References

12.1 Glossary of Terms & Acronyms

- **CO:** Course Outcome (SWE Curriculum educational milestone).
- **DFD:** Data Flow Diagram.
- **DND:** Do Not Disturb (Windows Focus Assist).
- **DTO:** Data Transfer Object.
- **FSM:** Finite State Machine.
- **HWND:** Win32 Window Handle.
- **IPC:** Inter-Process Communication.
- **LOC:** Lines of Code.
- **PID:** Process Identifier.
- **RLS:** Row-Level Security.
- **RTM:** Requirements Traceability Matrix.
- **UIA:** Windows UI Automation framework.

12.2 Key References (CO6)

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